Inflammatory bowel disease 1: investigation

NICE 2025, NG12, NICE 2019, NG129 & NG130, NICE 2013, DG11, NICE 2023, DG56, Gut 2022;71:1939, BMJ 2023;380:e068947, NICE 2018, The New Faecal Calprotectin Care Pathway, Gut 2019;68:s1, BMJ 2013;346:f432, JAMA 2023;330:951

Red Whale

GEMS

Guidelines & Evidence Made Simple

Symptoms and signs of inflammatory bowel disease

- Bowel symptoms: rectal bleeding, diarrhoea, increased stool frequency, urgency, tenesmus, rectal mass.
- Abdominal symptoms: pain, bloating, abdominal mass.
- Systemic symptoms: fever, fatigue, weight loss, sepsis.
- Bloods: raised inflammatory markers, anaemia.
- Nocturnal symptoms.

Many of these symptoms/signs cross over with red flag conditions, including bowel cancer and ovarian cancer, or other conditions, including irritable bowel syndrome, coeliac disease, endometriosis and hyperthyroidism.

Next steps include deciding how acutely someone needs to be seen, when and where.

Acutely unwell requiring admission? e.g. acute abdomen, acute severe inflammatory bowel disease, sepsis

Yes → **Admit that day**

No ↓

Could this be cancer? Incidence of colorectal cancer is increasing in younger patients

↓

Anal/rectal mass or anal ulceration

Yes → **Refer on suspected cancer pathway**

No ↓

Offer a FIT, and refer if ≥10mcgHb/g adults with ANY of:

- Abdominal mass
- Change in bowel habit
- Iron deficiency anaemia
- ≥60y with ANY anaemia, even in the absence of iron deficiency
- ≥40y with unexplained weight loss and abdominal pain
- <50y with rectal bleeding and either unexplained abdominal pain or weight loss
- ≥50y with ANY of: rectal bleeding OR abdominal pain OR weight loss

ACPGBI/BSG states to offer a FIT in <u>any</u> scenario where an adult has symptoms of possible colorectal cancer

FIT positive (≥10mcgHb/g) → **Refer on suspected cancer pathway**

None of the above/FIT negative ↓

Offer other tests to rule in IBD/rule out other conditions: stool culture (including *C. difficile*), FBC, CRP, coeliac serology, U&Es, bone profile, TFTs, CA125 in females (especially if ≥50y)

Abnormal → **Manage appropriately**

Normal ↓

Offer faecal calprotectin if 18–60y and IBS or IBD are suspected

- Calprotectin <100mcg/g → **IBD unlikely:** Make a positive diagnosis of IBS. Manage symptoms and safety-net
- Calprotectin 100–250mcg/g
- Calprotectin >250mcg/g

If taking NSAIDs, PPIs or aspirin, consider stopping this for 2w before repeat test

Calprotectin 100–250mcg/g → Repeat calprotectin within 2w

Calprotectin >250mcg/g → Clinical review

Clinical review: Stable or improving → Repeat calprotectin within 2w

Clinical review: Significant or worsening symptoms → **URGENT gastro referral**

Repeat calprotectin within 2w:

- <100 → **IBD unlikely:** Make a positive diagnosis of IBS. Manage symptoms and safety-net
- 100–250 → **ROUTINE gastro referral**
- >250 → **URGENT gastro referral**

IBD unlikely:
Make a positive diagnosis of IBS
Manage symptoms and safety-net

IF symptoms persist:

- Age <50y AND calprotectin <50 → **Try second-line IBS therapy before referral**
- Age ≥50y OR calprotectin ≥50 → **ROUTINE gastro referral**

In reality, we will likely arrange several tests at once to streamline care (FIT, stool culture, calprotectin, bloods, etc.). NICE is very clear though; a calprotectin is only for those where cancer is <u>not</u> suspected.

In secondary care, tests to make a formal diagnosis will be offered, e.g. colonoscopy, capsule endoscopy, CT and/or MRI.

2. Ulcerative colitis: management in adults

NICE 2019, NG130, Gut 2019;68:s1, JAMA 2023;330:951, BNF, accessed April 2025

Red Whale GEMS Guidelines & Evidence Made Simple

For every flare: inform secondary care and arrange inflammatory markers, FBC and stool for MC&S and *C. difficile*.
Patients may have a flare plan from secondary care: if so, follow it.

Management of initial episode/acute flare

Treatment aim: symptom control and mucosal healing

STOP! ANY symptoms or signs of acute severe disease

≥6 bloody stools/d AND ≥1 of:
- **T >37.8°C**
- **Pulse >90/min**
- **Haemoglobin <105g/L**
- **CRP or ESR >30mg/L**

OR severe pain/generally unwell

Yes → **Same-day admission**
For IV steroids/alternative treatment +/- colectomy

No ↓

Mild–moderate disease (mild: <4 bowel motions/d; moderate: 4–6 bowel motions/d and no severe features).

5-ASA: 5-aminosalycilate, e.g. sulfasalazine or mesalazine.
Oral 5-ASA can cause renal complications: check U&Es at baseline, after 2–3m and then annually.

Proctitis:	**Proctosigmoiditis/ left-sided ulcerative colitis**	**Extensive disease**
Topical 5-ASA (topical more effective than oral)	NICE: topical 5-ASA BSG: topical 5-ASA and oral 5-ASA	Offer topical 5-ASA AND high-dose oral 5-ASA

If no remission by 4 weeks

Proctitis:
- Consider differentials, e.g. infection (inc. STI), proximal constipation, Crohn's, IBS or rectal prolapse/ulcer
- ↓ Continue topical 5-ASA and ADD oral 5-ASA
- ↓ If further treatment needed, ADD topical or oral steroids

Proctosigmoiditis/left-sided ulcerative colitis:
- Consider high-dose oral 5-ASA AND topical 5-ASA (if not already taking) OR High-dose oral 5-ASA AND topical steroid
- ↓ If further treatment needed: Stop topical treatments. Offer oral 5-ASA with oral steroid

Extensive disease:
- Stop topical 5-ASA
- Use high-dose oral 5-ASA AND oral steroids

Any steroids (oral or topical) should be time limited (usually 4–8w maximum). Which steroids? BSG says: topical steroids – prednisolone suppository 5mg; oral steroids (in moderate UC) - prednisolone 40mg/d, wean over 6–8w; topical oral corticosteroids, e.g. budesonide MMX – can be used as alternative if wish to avoid systemic steroids.

If control still not achieved, contact secondary care: may admit and/or consider surgery or disease-modifying therapy.
When disease control achieved: move to maintenance treatment.

Maintenance treatment: for all with mild–moderate UC

Treatment aim: symptom control and mucosal healing

Proctitis/proctosigmoiditis:	**Left-sided UC or extensive disease**
Topical and/or oral 5-ASA (NICE prefers topical +/- oral; BSG says use topical for proctitis and, if more extensive disease, use oral)	Low maintenance dose of oral 5-ASA (to reduce the risk of colorectal cancer)

If oral 5-ASA appropriate, it can be more effective once daily (and improves adherence), but can cause more side-effects

5-ASA dosing for remission or maintenance

(Mild–moderate ulcerative colitis suitable for primary care treatment)

Topical 5-ASA	NICE: does not specify a dose and does not specify enema or suppository. BSG: for proctitis – 1g suppository, one at night; for more extensive disease – a 5-ASA enema. • Suppositories target the rectum (bedtime use gives best retention). Enemas act higher in the sigmoid. Several different topical 5-ASA preparations available: see BNF.
Oral 5-ASA	NICE: 'high dose' for inducing remission, and 'low maintenance dose' (no further specifics). BSG: to induce remission – start at 2–3g/d, increase to 4–4.8g/d if ongoing symptoms; for maintenance – use ≥2g/d.

If remission is not maintained with 5-ASA, OR if ≥2 exacerbations in 12m needing steroids, OR if cannot wean/stop steroids OR if 1 severe flare ever, secondary care may consider disease-modifying treatment.

3. Crohn's disease: management in adults

NICE 2019, NG129, Gut 2019;68:s1, BNF, accessed April 2025

Red Whale GEMS Guidelines & Evidence Made Simple

For every flare: inform secondary care and arrange inflammatory markers, FBC and stool for MC&S and *C. difficile*.
Patients may have a flare plan from secondary care: if so, follow it.

Management of 1st episode/1st exacerbation in 12m

Treatment aim: symptom control and mucosal healing

Offer steroid to induce remission

Systemically unwell — Yes → **Same-day admission**: For IV steroids, biological therapy and/or surgery

No ↓

STOP!

Think! Is this severe disease?

Very poor health AND ≥1 of:
- **≥3 diarrhoeal stools/d**
- **Weight loss**
- **Fever**
- **Severe abdominal pain**

Yes → NICE does not specify the site of care (primary/secondary care). We suggest having a low threshold for admission/contacting secondary care if concerned.

No ↓

Systemically well and primary care management appropriate

5-ASAs not routinely used in Crohn's

For active Crohn's colitis (mild, moderate or severe):
- Usually prednisolone. BSG recommends 40mg/d, tapering by 5mg/week for 8w.
- If extensive disease/poor prognostic features, early biological therapy may be advised by secondary care.

For distal ileal, ileocaecal or right-sided mild–moderate (not severe) disease:
- If prednisolone contraindicated/not tolerated:
  - Consider budesonide (less effective than prednisolone, but potentially fewer side-effects) (NICE).
- If mild–moderate ileocaecal Crohn's disease, BSG recommends *ileal release* budesonide MR (9mg once daily) for 8w to induce remission; once remission achieved, taper over 2–4w.
- Note: if distal ileal Crohn's only, surgery may be considered early in disease course.

Enteral nutrition (secondary care decision):
- An alternative to steroids for induction of remission in children/young people who are growing.
- BSG suggests it can also be used in adults who want to avoid steroids, if motivated to adhere for 8w.

If steroids contraindicated/not tolerated and mild–moderate disease, NICE suggests considering aminosalicylate. BSG recommends against aminosalicylate use in induction or maintenance of Crohn's: specialist decision we think!

≥2 inflammatory exacerbations/year OR unable to taper and reduce steroids

Secondary care management:
- Azathioprine, mercaptopurine or methotrexate may be added to steroids to induce remission.
- The MABs, e.g. infliximab or adalimumab, may be considered for severe/unresponsive disease.

Maintenance

Treatment aim: symptom control and mucosal healing

For ALL with Crohn's:
- Advise against smoking (stopping smoking significantly reduces relapse rate).
- Colorectal cancer surveillance: see page 4 of the GEMS for specifics.

Maintenance treatment: not for all; consider risks of flare and inflammation vs. drug side-effects.

No maintenance treatment	Maintenance treatment
Review promptly if relapse occurs	Steroids should NOT be used for maintenance Treatment may be with azathioprine, mercaptopurine, methotrexate or a biological

Surgery

Surgery will be required in 50–80% with Crohn's at some point for complications including strictures and fistulae.

4. IBD in adults: microscopic colitis and IBD complications

BJGP2021;71:41, BNF accessed April 2025, Gut 2019;68:s1, NICE 2022, CG118, Diseases 2022;11:6

Red Whale
GEMS
Guidelines & Evidence Made Simple

Microscopic colitis: management (as with other types, admit if patient is systemically unwell)

Steroids to induce remission (be guided by secondary care)

Oral budesonide 9mg/d for 6–8w, then taper over 2–4w.

Budesonide is sometimes needed for longer, e.g. ≥6m at 3–6mg/d.

Remember adrenal risks: taper steroids and don't stop abruptly.

If poor response to steroids, secondary care may suggest azathioprine, mercaptopurine or a biological agent.

Do not use prednisolone, loperamide, bismuth, mesalazine, antibiotics or probiotics.

Medication review

Microscopic colitis is associated with drugs including aspirin, PPIs, NSAIDs, SSRIs and statins.

If on these, can any be changed?

COMPLICATIONS of IBD (all types)

IBD has wider impact beyond the bowel; how can we best support our patients in primary care?

Anaemia	Iron deficiency anaemia occurs in 1/3 with active IBD. Colorectal cancer is also a cause. Non-iron deficiency anaemia may occur, e.g. B12/folate deficiency, anaemia of chronic disease. Ferritin is unreliable in inflammatory conditions: ferritin up to 100 can signify iron deficiency in inflammation; transferrin saturation may help. Systemic inflammation inhibits oral iron absorption; iron tablets shouldn't be used in active disease. IV iron is first line if intolerant to oral iron, active IBD and Hb <100g/L, or in those needing erythropoietin. If oral iron used, use once-daily or alternate-day dosing (do not give >100mg elemental iron/d). What to do in practice? Does the patient meet the criteria for a suspected cancer pathway referral or a FIT? Yes: refer/investigate. No: assess for underlying cause, treat and liaise with secondary care.
Osteoporosis	Affects 15% with IBD (and 35–40% will have osteopenia). Factors that increase the risk: • Prolonged/high-dose oral steroid use (>7.5mg/d prednisolone for ≥3m is significant). • Uncontrolled inflammation. • Malabsorption (especially in Crohn's disease) and weight loss. • Lack of physical activity. Advise about stopping smoking, reducing alcohol excess and doing regular weight-bearing exercise. *See Osteoporosis article for more details.*
Vitamin D and calcium	Over 50% of those with IBD are vitamin D deficient; check vitamin D in all and supplement if low. If taking oral steroids for a flare, BSG advises 800 IU/d vitamin D and 800–1000mg/d calcium, unless dietary calcium intake is adequate (use a calcium calculator to check).
Malnutrition	Assess BMI and ask about weight loss; up to 85% of patients with IBD are under-nourished. All at risk of malnutrition should be referred to a dietician. All with IBD who are malnourished/at risk of malnutrition should have screening bloods for nutritional deficiencies (iron stores, vitamin B12, folate, vitamins A, C, D and E, potassium, calcium, magnesium, phosphate, zinc and selenium).
Colorectal cancer risk and surveillance	Increased risk of colorectal cancer. Risk increases with severity of inflammation and time from diagnosis. For both UC and Crohn's disease, risk is estimated at 2% after 10y, 8% after 20y and 18% after 30y. In ulcerative colitis, some medications have a chemoprophylactic effect: • Left-sided/more extensive disease: daily oral 5-ASA reduces the risk of colorectal cancer. • Secondary care may advise thiopurines to reduce the risk of colorectal cancer. Colonoscopic surveillance: BSG and NICE differ in their guidance: • BSG: if colonic disease, start after 8y, or from diagnosis if primary sclerosing cholangitis. • NICE: start after 10y if UC (but not proctitis alone) or if Crohn's involving >1 segment of colon. When surveillance should start/frequency of ongoing surveillance will be guided by secondary care.

5. IBD: beyond the bowel

Gut 2019;68:s1, NICE 2022, CG118, The Green Book, accessed February 2025, NICE 2017, CG146, www.crohnsandcolitis.org.uk., BMJ 2013;346:f432, JAMA 2023;330:951

Red Whale
GEMS
Guidelines & Evidence Made Simple

IBD has wider impact beyond the bowel; how can we best support our patients in primary care?

IBD: associated symptoms and quality of life

Pain	Pain in IBD is complex. It can be due to physical or psychological factors, or both: • Rule out pain due to IBD, e.g. active disease, stricture, abscess or adhesions. • Consider co-existing conditions, e.g. IBS or fibromyalgia. Psychological interventions can help for unexplained chronic pain in IBD. Opioids: avoid long-term use – associated with poor outcomes. NSAIDs: Short-term: consider if IBD in remission, but use is associated with a 20% relapse rate. Long-term: associated with higher disease activity scores (esp. in Crohn's), but not relapse.
Fatigue	Fatigue is common, can be multifactorial and doesn't necessarily correlate with disease activity. Investigate for subclinical disease/other modifiable risk factors (e.g. poor sleep, medication review). Check bloods, including FBC, haematinics, U&Es, TFTs, B12 and vitamin D. If no correctable cause/active IBD, psychotherapy, stress management or graded exercise may help.
Smoking	Crohn's: smoking is associated with increased risk of flares, surgery and worse surgical outcomes. Ulcerative colitis: smoking is possibly protective, but is outweighed by the health risks of smoking! There is an increased risk of a UC flare on stopping smoking.
Cervical cancer	Cohort data has shown an association between IBD and cervical dysplasia, mainly in those on oral immunosuppressive treatment or those who smoke. Encourage cervical screening.
Sex and body image	Stomas, pain or a fistula may all impact on sex life and body image. We can help with these conversations in primary care. Crohn's and Colitis UK has fantastic downloadable PDFs for patients, with lots of tips.
Psychological wellbeing	Depression and anxiety are commoner in IBD. Stress can increase the risk of a flare. Ask about psychological wellbeing. Treatment is the same as for the general population. Fear of incontinence/inability to find a toilet can contribute to anxiety and social stress. A RADAR key or Can't Wait card can help. Details can be found on Crohn's and Colitis UK website.
Travel	Travel concerns can range from dietary precautions to managing medications; from dealing with diarrhoea to travel insurance. The IBD Passport website has loads of useful information for patients. For those on immunosuppression, BSG says to have a low threshold for treating possible travellers' diarrhoea with either a quinolone or azithromycin. The BSG guidance predated the MHRA guidance on the risks of quinolones (see the articles *Travellers' diarrhoea* and *Quinolones (the -floxacins): the risks* on Red Whale Knowledge).
Vaccines	Biological agents, DMARDs and steroids cause immunosuppression to differing degrees. • Significant immunosuppression is a <u>contraindication to a live vaccine</u>. • Those who are significantly immunosuppressed require additional pneumococcal vaccination. Check The Green Book for specifics. Infants exposed to biologics in utero should not receive live vaccines for 6m after birth (rotavirus, BCG).
Autoimmune conditions	27–30% of those with UC have autoimmune inflammation of other organs: • Liver: autoimmune liver disease, primary sclerosing cholangitis. • Joints: seronegative arthropathy, ankylosing spondylitis, sacroiliitis. • Eye: scleritis, episcleritis, anterior uveitis. • Skin: erythema nodosum, pyoderma gangrenosum.